

MODULE-4:

AWS Cloud Security Module Overview

This document details the AWS Academy Cloud Foundations Module 4: AWS Cloud Security. The module's main goal is to introduce the AWS approach to security, covering both AWS's internal controls and the products and features customers use to achieve their security objectives. The module includes an educator-led activity, a recorded demonstration, and a hands-on lab, culminating in a knowledge check.

Key Module Topics and Objectives:

The module covers these topics:

- The AWS shared responsibility model
- AWS Identity and Access Management (IAM)
- Securing a new AWS account
- Securing accounts (using AWS Organizations)
- Securing data on AWS (encryption at rest and in transit)
- Ensuring compliance (AWS compliance programs, AWS Config, AWS Artifact)
- Additional security services and resources (AWS KMS, Amazon Cognito, AWS Shield)

After completing the module, learners should be able to:

- Understand the shared responsibility model and identify AWS and customer responsibilities.
- Recognize IAM users, groups, and roles, and describe different security credentials.
- Identify steps for securing a new AWS account.
- Understand how to secure AWS data.
- Recognize AWS compliance programs.

Shared Responsibility Model:

The AWS shared responsibility model divides security responsibilities between AWS and the customer.

- **AWS Responsibility (Security "of" the Cloud):** AWS manages the underlying infrastructure, including physical security, hardware, software, networking, and facilities. This includes data center physical security, access control, intrusion detection, and virtualization infrastructure.
- **Customer Responsibility (Security "in" the Cloud):** Customers are responsible for securing the services and applications they deploy on AWS. This includes operating system management (patches, updates), application security, security group configuration, network configurations, and account management. The level of customer responsibility varies depending on the service model (IaaS, PaaS, SaaS).

- **IaaS (Infrastructure as a Service):** Customers have more control and responsibility for security configuration and management. Examples include Amazon EC2, Amazon EBS, and Amazon VPC.
- **PaaS (Platform as a Service):** AWS manages more of the underlying infrastructure, but customers remain responsible for data security and access control. Examples include Amazon RDS and AWS Lambda.
- **SaaS (Software as a Service):** AWS manages most security aspects. Examples include AWS Trusted Advisor, AWS Shield, and Amazon Chime.

Scenarios illustrating AWS vs. customer responsibilities are provided in the document with answers. Key takeaways from Section 1 emphasize the shared responsibility and AWS's responsibility for the underlying infrastructure.

AWS Identity and Access Management (IAM):

IAM is a free AWS service to manage access to AWS resources. Key components include:

- **IAM Users:** Individuals or applications that authenticate with an AWS account. They require unique names and security credentials.
- **IAM Groups:** Collections of users with identical authorizations. They simplify permissions management.
- **IAM Policies:** Documents defining resource access permissions (allow and deny statements). They can be identity-based (attached to users, groups, or roles) or resource-based (attached to resources). Explicit deny statements always override allow statements. The IAM Policy Simulator is a helpful tool for testing and troubleshooting policies.
- **IAM Roles:** Mechanisms for granting temporary access to AWS resources without long-term credentials. They're useful for applications, services, or cross-account access.

IAM authentication involves proving identity, and authorization determines what actions are permitted. MFA (multi-factor authentication) is strongly recommended for enhanced security. Section 2 key takeaways summarize IAM components and policies. A recorded IAM demo is referenced.

Securing a New AWS Account:

Best practices for securing a new AWS account include:

1. **Stop using the root user:** Create an IAM user with appropriate permissions and disable the root user access keys.
2. **Enable MFA:** Require MFA for the root user and all IAM users.
3. **Use AWS CloudTrail:** Enable CloudTrail to track user activity and API requests.
4. **Enable billing reports:** Utilize the AWS Cost and Usage Report to monitor spending.

Key takeaways from Section 3 highlight these best practices.

Securing Accounts (AWS Organizations):

AWS Organizations allows centralized management of multiple AWS accounts. Security features include:

- Grouping accounts into OUs (Organizational Units) and applying different access policies.
- IAM integration: Permissions are the intersection of AWS Organizations policies and IAM permissions.
- SCPs (Service Control Policies): Control maximum permissions for member accounts, overriding account administrators.

Securing Data on AWS:

This section emphasizes data encryption:

- **Encryption at rest:** Encoding data stored physically (disk, tape) using AWS KMS. Supported services include Amazon S3, Amazon EBS, Amazon EFS, and Amazon RDS managed databases.
- **Encryption in transit:** Using TLS/SSL to secure data moving across a network. AWS Certificate Manager helps manage certificates.

Securing Amazon S3 buckets and objects involves managing access control:

- Newly created buckets are private by default.
- Options for controlling access include Amazon S3 Block Public Access, IAM policies, bucket policies, ACLs, and AWS Trusted Advisor checks.

Working to Ensure Compliance:

AWS offers various compliance programs, certifications, and tools:

- **AWS Compliance Programs:** Information on AWS policies and controls.
- **AWS Config:** Assessing, auditing, and evaluating AWS resource configurations.
- **AWS Artifact:** Access to compliance-related documents and reports.

Key takeaways from Section 6 summarize these tools and programs.

Additional Resources:

The document concludes with links to additional resources on AWS security. A sample exam question and answer are also included. The module wraps up with a summary and instructions to complete a knowledge check.